

Dummit & Foote 7.3

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1 Problems

When we discuss the ideal generated by a set $A \subset R$, we require $1 \in R$ so that $A \subset RA$. RA is the left ideal generated by A , AR is the right ideal generated by A , RAR is the ideal generated by A . When R is commutative, $RA = AR = RAR = (A)$.

The trivial ideal and R are both principal: $0 = (0), R = (1)$.

Let R have $1 \in R$ and $I \triangleleft R$. Then $I = R$ iff I has a unit.

Suppose $u \in I$. Then $1 = u^{-1}u \in I$, so $r \in I$, or $R = I$. If $R = I$, $1 \in I$.

Let R be commutative such that its only ideals are 0 and R . Then R is a field.

Let $u \neq 0$ in R . Then $(u) = I$, so there exists $uv = vu = 1$. That is, every $u \neq 0$ is a unit.

Definition: Let $M \triangleleft S$. M is a maximal ideal if $M \neq S$ and the only ideals containing M are M and S .

Prop: Let R be a ring with identity. Then any proper ideal I is contained in a maximal ideal.

Proof: Let $S = \{J \triangleleft R : I \subset J, J \neq R\}$. S is a nonempty poset ordered by inclusion. Let $\mathcal{I}$ be a set of indices for which $I_{\mathcal{I}}$ is a totally ordered set of ideals I_i . Take $K = \cup_{i \in \mathcal{I}} I_i$. $K \triangleleft R$ is an ideal. Since each I_i contains no units, K contains no units. Hence, K is a proper ideal, so $K \in S$. That is, each chain has an upper bound. By Zorn's Lemma, I is contained in a maximum ideal. By convention sets of subrings use `mathcal`.

Problem: Let $N \triangleleft G$. Show that any subgroup K of G/N is of the form H/N , where $N \subset H \subset G$.

Proof: Let $S \subset G$ be such that $sN \in K$ for all $s \in S$. Note $(sN)(s^{-1}N) = N$ and $N \in K$, so $s^{-1}N \in K$ as well. S is closed under inverses. $(sN)(tN) = stN$, so S is closed under multiplication. Also, $nN = N$ for all $n \in N$, so $N < S$. S is our H that we are looking for, AWD.

Prop 12: Let R be commutative. $M \triangleleft R$ is maximal iff R/M is a field.

Proof: The ideals of R containing M are in bijection with the ideals of R/M by the lattice theorem. M is maximal iff R/M only contains the ideals R/M and 0 , which happens iff R/M is a field.

Method to construct fields: take a commutative ring with identity R and some maximal ideal I and take R/I .

Prop 13: Assume R is commutative. Then the ideal P is prime iff R/P is an integral domain.

Suppose P is a prime ideal and $ab + P = (a + P)(b + P) = 0 + P$, or that $ab \in P$. Then either $a \in P$ or $b \in P$, for which $a + P = P$ or $b + P = P$. Hence, R/P is an integral domain. Now for the reverse direction, $ab \in P$ implying $a + P = P$ or $b + P = P$ is the same as the definition of a prime ideal.

Corollary 14: Let R be commutative and $M \triangleleft R$. Then M is a prime ideal.

Proof: R/M is a field, which is also an integral domain. By prop 13, M is a prime ideal.

P 2. Assume R is commutative with identity 1. Show that the augmentation ideal of the group ring RG is generated by $S = \{g - 1 : g \in G\}$. If $G = \langle \sigma \rangle$ is cyclic, the augmentation ideal is generated by $\sigma - 1$.

The elements of the augmentation ideal I are of the form $\sum_{i=1}^n r_i g_i$, where $\sum_{i=1}^n r_i = 0$. Hence $r_1 = -r_2 - r_3 - \dots - r_n$, so an arbitrary element of I is of the form

$$r_2(g_2 - 1) + r_3(g_3 - 1) + \dots + r_n(g_n - 1).$$

We have $I \subset (S)$. Moreover, $S \subset I$, so $(S) \subset I$. Hence $I = (S)$.

If G is cyclic, $g^k - 1 = \sum_{i=0}^{k-1} g^i(g - 1) \in (g - 1)$. Hence, the augmentation ideal is a principal ideal in this case.

P 3. (a) Let p be a prime and G be an Abelian p -group. Show that the Nilradical of $\mathbb{F}_p G$ is also the augmentation ideal.

Let $\varphi : RG \rightarrow R$ be the ring homomorphism such that $\varphi(x)$ is the sum of the R -coefficients of x . An element of the Nilradical of RG is such that

$\varphi(x^n) = \varphi(0) = 0$, so $\varphi(x)^n = 0$. R is a field, so $\varphi(x) = 0$. That is, x is in the augmentation ideal. Conversely, the Augmentation ideal is generated by elements of the form $g - 1$. By the Binomial theorem, $(g - 1)^{p^m} = g^{p^m} - 1$ (still works in $p = 2$). We take $|G| = p^m$, AWD. Each element of the Augmentation ideal is in the Nilradical.

(b) It suffices to show $(g - 1)(g_1 + g_2 + \cdots + g_n) = 0$. This expands as

$$gg_1 + gg_2 + \cdots + gg_n - g_1 - g_2 - \cdots - g_n = 0,$$

since multiplication by g permutes G .

P 5. Prove that if M is an ideal such that R/M is a field, then M is maximal.

Since R/M is a field, its only ideals are 0 and itself. By the Lattice theorem, the only ideals of R containing M are M and R . The result follows from the definition of maximality.

P 8. Let R be an integral domain. Prove that $(a) = (b)$ for some $a, b \in R$ iff $a = ub$ for some unit u .

Suppose either of a, b are 0. Then both ideals are 0, for which both a, b are 0. In this case, $0 = 1 \cdot 0$. Else, $a = ub$, $b = va$ for some $u, v \in R$. Then $ab = abuv$, or $ab(uv - 1) = 0$. $ab \neq 0$, so $uv = 1$. Hence, u is indeed a unit, AWD.

P 9. Let $a \in [0, 1]$ and let I_a denote the set of functions f in R such that $f(a) = 0$. If $f, g \in I_a$, $f - g \in I_a$, so I_a is a subring. $(fg)(a) = (gf)(a) = 0$, so I_a is an ideal. $I = I_{1/3} \cap I_{1/2}$ gives the ideal in question. This is not a prime ideal because $f(x) = (x - 1/3)(x - 1/2) \in I$, but neither $x - 1/3, x - 1/2$ are in I .

P 11. Let R be commutative and I, J be ideals, P be a prime ideal containing IJ . Show that P contains either I or J .

Suppose FTSOC there exists $a \in I$ and $b \in J$ such that $a, b \notin P$. Then $ab \in IJ \in P$, so either $a \in P$ or $b \in P$, a contradiction.

P 19. Let R be a finite commutative ring. Show that every prime ideal is also maximal.

Let $P \triangleleft R$ be prime. Then P/R is a finite integral domain, which is also a field F ($R \neq P$). Hence, P is a maximal ideal.

P 20. Prove that a nonzero finite commutative ring with no zero divisors is a field.

Let $r \in R$ not be a zero divisor. r permutes the elements of R . Since R is finite, r can be encoded as an element of S_n , where $n = |R|$. Hence, there exists an integer m such that r^m acts as the identity, or $r^m = 1$. Hence R is an integral domain, from corollary 3 we conclude that R is a field.

Q ?. If R is a finite ring with an element not a zero divisor, then R has identity.

Proof: Let $r \in R$ not be a zero divisor. r permutes R by left multiplication, and so does r^m . Take m such that r^m acts as the identity permutation. Then R has a left identity 1_L . Similarly, R has a right identity 1_R . Since $1_R = 1_L 1_R = 1_L$, this identity is unique.

P 21?. Prove that a finite ring with no zero divisors with identity $1 \neq 0$ that has no zero divisors is a field.

Q ?. Let R be a finite ring with no zero divisors, $|R| > 1$. Then R is a field.

P 22. (a) Prove that the $\mathbb{F}_p$ Quaternions are a homomorphic image of the integer Quaternions.

Take the homomorphism to be the coefficients of the integer quaternions modulo p .

(b) Prove that the $\mathbb{F}_p$ Quaternions contain zero divisors.

By Legendre's Four-square theorem, there exists integers a, b, c, d for which $a^2 + b^2 + c^2 + d^2 = p$. That is,

$$N(a + bi + ci + di) = (a + bi + cj + dk)(a - bi - cj - dk) = p = 0.$$

a, b, c, d cannot all be zero in $\mathbb{F}_p$, so the $\mathbb{F}_p$ Quaternions contain zero divisors.

P 23. Show that every prime ideal of a boolean ring is also maximal.

Let P be a prime ideal of boolean ring B . B/P is an ntegral domain. Let $x \in B/P$ be nonzero. Then $xx = x$, so for any $b \in B$, $bx = bx$, or $bx = b$ for all $b \in B$. Similarly, $xb = b$ for all $b \in B$. All nonzero elements in B/P are the identity, so $B/P \cong \mathbb{F}_2$, from which P is maximal.

P 25. Let R be a commutative ring and assume for every $a \in R$, there exists $n > 1$ such that $a^n = a$. Show that every prime ideal P of R is also maximal.

Let $a \in R/P$ be nonzero. $ra^n = ra$ for all $r \in R/P$. Since a is not a zero divisor, $ra^{n-1} = r$ for all $r \in R/P$. That is, a^{n-1} is a right identity. Similarly, a^{n-1} is a left identity, so $a^{n-1} = 1$. If $n = 2$, a is the identity, which has an inverse. Else, a^{n-2} is the inverse of a . Every nonzero element of commutative

ring with identity R/P is a unit, so R/P is a field. Hence, P is maximal.

P 26. Show that the Nilradical of a commutative ring R is contained in the intersection of all prime ideals of R .

Let P be a prime ideal. R/P is an ntegral domain, so for any Nilpotent n , $(n + P)^m = P$, or $n \in P$. That is, $N(R) \subset P$.

P 28. Let R be a commutative ring and $N = (a_1, a_2, \dots, a_m)$, where each a_i is Nilpotent. Show that N is a Nilpotent ideal. As a consequence, if the Nilradical of R is finitely generated, it is Nilpotent.

Each a_i has an associated n_i for which $a_i^{n_i} = 0$. Let $n = n_1 + n_2 + \dots + n_m$. I claim $N^n = 0$. Each element of N^n is a sum of products of n elements of N . By Pigeonhole, each term contains at least e_j copies of some a_j where $e_j \geq n_j$. That is, each term is zero, making $N^n = 0$.

P 30. Let I be an ideal of a commutative ring R . Let

$$\text{rad } I = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{Z}^+\}.$$

Prove that $\text{rad } I$ is an ideal containing I and that the $\text{rad } I/I$ is the Nilradical of the quotient ring R/I .

$\text{rad } I$ contains 0, so is nonempty. Let m, n be st $r^m, s^n \in I$. Since R is commutative $(r - s)^{m+n}$ may be expanded by binomial theorem. Each term contains either $r^k \in I$ or $s^j \in I$, where $k \geq m$ and $j \geq n$, for which the term will be in I . $\text{rad } I \triangleleft R$. Hence, $(r - s)^{m+n} \in I$, or $r - s \in \text{rad } I$.

$\text{rad } I$ is closed under left multiplication: $(xr)^n = x^n r^n \in I$. R is commutative, so right closure follows. Hence, $\text{rad } I \triangleleft R$. Take $n = 1$, so $I \subset \text{rad } I$.

$N(R/I) = \{\bar{r} : \bar{r}^n = \bar{0}\} = \text{rad } I/I$. Alternatively, $N(R/I) \triangleleft R/I$. By the Lattice theorem, $\text{rad } I \triangleleft R$.

P 31. Let R be a commutative ring. An ideal I is called a radical ideal if $\text{rad } I = I$. (a) Every prime ideal is radical.

$N(R/P) = \text{rad } I/I$. Every nilpotent element is either 0 or a zero divisor. Since R/P is an integral domain, $N(R/P)$ is the zero ring. This forces $\text{rad } I = I$.

P 35. Let $\mathcal{S}$ denote the poset of ideals that do not contain A . $\mathcal{S}$ is nonempty since $0 \in \mathcal{S}$. Let $\mathcal{C}$ be a chain of ideals not containing A . Consider the ideal $J = \cup_{I \in \mathcal{C}} I$. Suppose FTSOC $A \subset J$. For all $i \in [n]$, there exists index j such that $a_i \in I_j$. Let k be such that I_k contains each I_j . Then $A \subset I_k \subset J$, contradiction, so $J \in \mathcal{S}$. Each chain has a maximal element, so by Zorn's Lemma, $\mathcal{S}$ has a maximal element.

P 37. Let R be a commutative ring with identity. A local ring R is a ring with a unique maximal ideal M . Show that every element of $R - M$ is a unit.

Let U denote the set of units. Let $x \in R - U$. Note that $(x) \triangleleft R$ is a proper ideal. Otherwise, $1 \in (x)$, or $rx = xr = 1$ for some $r \in R$, contradiction. We have $(x) \subset M$, so $x \in M$. The contrapositive is if $x \notin M$, then $x \in U$. Hence, $R - M \subset U$. On the other hand, M cannot contain any units as it would otherwise not be proper, so $U \subset R - M$. Hence $U = R - M$.

Converse: Let R be commutative with identity such that the non-unital elements form an ideal M . Show that R is local and that M is the unique maximal ideal. Any ideal I is contained in $M = R - U$, as otherwise $I = R$. Let x be nonzero in R/M . Then $x \in U$, for which we may find y such that $xy = 1$. Every nonzero element of commutative ring R/M is a unit, so R/M is a field and M is maximal.

P 40. ????????????